THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 225.5 m²	Canvas = CSA of cylinder + CSA of cone; $= 88 + 137.5 = 225.5$ m²
2	**(a)** 396 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(6^3 - 3^3)$; Volume $= 396$ cm³
3	**(c)** 6835.5 cm³	Cube $= 9261$ cm³; hemisphere of radius 10.5 cm $= \frac{2}{3}\pi r^3 = 2425.5$ cm³; Volume $= 6835.5$ cm³
4	**(c)** 256 cm	$\frac{4}{3}\pi(12)^3 = \pi(3)^2 L$; $L = 256$ cm
5	**(c)** 1 cm	$\frac{4}{3}\pi(3)^3 = \pi(6)^2 x$; $x = 1$ cm
6	**(a)** 1056 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 10 + 4 \times \frac{22}{7} \times 7^2$; TSA $= 1056$ cm²
7	**(b)** 20328 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 42 \times 70 + 2 \times \frac{22}{7} \times 42^2 = 20328$ cm²
8	**(b)** 192	$n = \frac{24 \times 12 \times 18}{3^3}$; $n = 192$
9	**(a)** 357.5 m²	Canvas = CSA of cylinder + CSA of cone; $= 220 + 137.5 = 357.5$ m²
10	**(d)** 1056 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(8^3 - 2^3)$; Volume $= 1056$ cm³

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of sphere = volume of cone; $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(6)^2(3)$ gives $R^3 = 27$ (1); $R = 3$ cm (1).	1+1
12	Volume of cylinder $= \pi \times 6^2 \times 48$; Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 48$ (1).	1+1
13	TSA = surface of cube − circle covered + curved surface of hemisphere; $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 17.5$ cm (1); TSA $= 7350 + 962.5 = 8312.5$ cm² (1).	1+1
14	Volume of sphere = volume of cone; $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(3)^2(96)$ gives $R^3 = 216$ (1); $R = 6$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 4158$ cm³; two hemispheres $= 4851$ cm³ (1); Volume $= 9009$ cm³ (1).	1+1+1
16	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 6468$ cm³ (1); Volume $= 25872$ cm³ (1).	1+1+1
17	Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 3.5$, $h = 15$ (1); TSA $= 330 + 154 = 484$ cm² (1).	1+1+1
	OR: Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 6237$ cm³; two hemispheres $= 4851$ cm³ (1); Volume $= 11088$ cm³ (1).	1+1+1
18	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 19404$ cm³ (1); Volume $= 38808$ cm³ (1).	1+1+1
	OR: Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 7$, $h = 10$ (1); TSA $= 440 + 616 = 1056$ cm² (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(20) = \pi(4)^2 x$ (2); $x = 15$ cm (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 3080$ cm³ and base area of cuboid $= 308$ cm² (2); Height $= 3080 \div 308 = 10$ cm (1).	2+2+1
20	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(16) = \pi(4)^2 x$ (2); $x = 12$ cm (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 4620$ cm³ and base area of cuboid $= 231$ cm² (2); Height $= 4620 \div 231 = 20$ cm (1).	2+2+1

Mistakes to avoid:

- Counting the joined faces in the total surface area of a combination of solids (e.g. adding the base of a cone resting on a hemisphere).
- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
- Using diameter in place of radius while substituting into a formula.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 54684 cm³	Cube $= 74088$ cm^3; hemisphere of radius 21 cm $= \frac{2}{3}\pi r^3 = 19404$ cm^3; Volume $= 54684$ cm^3
2	**(d)** 243 cm	$\frac{4}{3}\pi(9)^3 = \pi(2)^2 L$; $L = 243$ cm
3	**(a)** 2 cm	$\frac{4}{3}\pi(6)^3 = \pi(12)^2 x$; $x = 2$ cm
4	**(b)** 3344 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 14 \times 10 + 4 \times \frac{22}{7} \times 14^2$; TSA $= 3344$ cm^2
5	**(a)** 13728 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 28 \times 100 + 2 \times \frac{22}{7} \times 28^2 = 13728$ cm^2
6	**(b)** 240	$n = \frac{24 \times 18 \times 15}{3^3}$; $n = 240$
7	**(c)** 902 m²	Canvas = CSA of cylinder + CSA of cone; $= 352 + 550 = 902$ m^2
8	**(a)** 3168 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(12^3 - 6^3)$; Volume $= 3168$ cm^3
9	**(d)** 184558.5 cm³	Cube $= 250047$ cm^3; hemisphere of radius 31.5 cm $= \frac{2}{3}\pi r^3 = 65488.5$ cm^3; Volume $= 184558.5$ cm^3
10	**(a)** 18 cm	$\frac{4}{3}\pi(6)^3 = \pi(4)^2 L$; $L = 18$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of cylinder $= \pi \times 6^2 \times 72$; Volume of a sphere $= \frac{4}{3}\pi(2)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 243$ (1).	1+1
12	TSA = surface of cube − circle covered + curved surface of hemisphere; $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 24.5$ cm (1); TSA $= 14406 + 1886.5 = 16292.5$ cm^2 (1).	1+1
13	Volume of sphere = volume of cone; $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(12)^2(48)$ gives $R^3 = 1728$ (1); $R = 12$ cm (1).	1+1
14	Volume of cylinder $= \pi \times 9^2 \times 32$; Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 72$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 16632$ cm^3; two hemispheres $= 38808$ cm^3 (1); Volume $= 55440$ cm^3 (1).	1+1+1
16	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 808.5$ cm^3 (1); Volume $= 3234$ cm^3 (1).	1+1+1
17	Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 7$, $h = 12$ (1); TSA $= 528 + 616 = 1144$ cm^2 (1).	1+1+1
	OR: Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 33264$ cm^3; two hemispheres $= 38808$ cm^3 (1); Volume $= 72072$ cm^3 (1).	1+1+1
18	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 4042.5$ cm^3 (1); Volume $= 6468$ cm^3 (1).	1+1+1
	OR: Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 14$, $h = 15$ (1); TSA $= 1320 + 2464 = 3784$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(12) = \pi(6)^2 x$ (2); $x = 4$ cm (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 1848$ cm^3 and base area of cuboid $= 462$ cm^2 (2); Height $= 1848 \div 462 = 4$ cm (1).	2+2+1
20	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(18) = \pi(6)^2 x$ (2); $x = 6$ cm (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 6160$ cm^3 and base area of cuboid $= 308$ cm^2 (2); Height $= 6160 \div 308 = 20$ cm (1).	2+2+1

Mistakes to avoid:

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- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 27 cm	$\frac{4}{3}\pi(9)^3 = \pi(6)^2 x$; $x = 27$ cm
2	**(d)** 7392 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 21 \times 14 + 4 \times \frac{22}{7} \times 21^2$; TSA $= 7392$ cm²
3	**(d)** 17710 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 35 \times 91 + 2 \times \frac{22}{7} \times 35^2 = 17710$ cm²
4	**(b)** 256	$n = \frac{24 \times 24 \times 12}{3^3}$; $n = 256$
5	**(d)** 1078 m²	Canvas = CSA of cylinder + CSA of cone; $= 528 + 550 = 1078$ m²
6	**(d)** 3564 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(12^3 - 3^3)$; Volume $= 3564$ cm³
7	**(a)** 437472 cm³	Cube $= 592704$ cm³; hemisphere of radius 42 cm $= \frac{2}{3}\pi r^3 = 155232$ cm³; Volume $= 437472$ cm³
8	**(b)** 144 cm	$\frac{4}{3}\pi(12)^3 = \pi(4)^2 L$; $L = 144$ cm
9	**(a)** 16 cm	$\frac{4}{3}\pi(12)^3 = \pi(12)^2 x$; $x = 16$ cm
10	**(b)** 594 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 3.5 \times 20 + 4 \times \frac{22}{7} \times 3.5^2$; TSA $= 594$ cm²

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	TSA = surface of cube − circle covered + curved surface of hemisphere; $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 28$ cm (1); TSA $= 18816 + 2464 = 21280$ cm² (1).	1+1
12	Volume of sphere = volume of cone; $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(18)^2(9)$ gives $R^3 = 729$ (1); $R = 9$ cm (1).	1+1
13	Volume of cylinder $= \pi \times 12^2 \times 24$; Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 96$ (1).	1+1
14	TSA = surface of cube − circle covered + curved surface of hemisphere; $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 31.5$ cm (1); TSA $= 23814 + 3118.5 = 26932.5$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 8662.5$ cm³; two hemispheres $= 4851$ cm³ (1); Volume $= 13513.5$ cm³ (1).	1+1+1
16	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 9702$ cm³ (1); Volume $= 29106$ cm³ (1).	1+1+1
17	Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 10.5$, $h = 20$ (1); TSA $= 1320 + 1386 = 2706$ cm² (1).	1+1+1
	OR: Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 11088$ cm³; two hemispheres $= 38808$ cm³ (1); Volume $= 49896$ cm³ (1).	1+1+1
18	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm³; cone $= \frac{1}{3}\pi r^2 h = 4851$ cm³ (1); Volume $= 7276.5$ cm³ (1).	1+1+1
	OR: Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 14$, $h = 25$ (1); TSA $= 2200 + 2464 = 4664$ cm² (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(9)^2(8) = \pi(3)^2 x$ (2); $x = 24$ cm (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 4620$ cm³ and base area of cuboid $= 660$ cm² (2); Height $= 4620 \div 660 = 7$ cm (1).	2+2+1
20	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(9)^2(16) = \pi(6)^2 x$ (2); $x = 12$ cm (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 2310$ cm³ and base area of cuboid $= 924$ cm² (2); Height $= 2310 \div 924 = 2.5$ cm (1).	2+2+1

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- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.
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